Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st part–June 5th 2025

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and ensure you provide a brief explanation. Remember, without a reason, the answer will be considered invalid.

A. How does the PRAM model abstract away synchronization and communication, and what are the implications for algorithm design? (1)

B. How do the different types of concurrent write (CW) models affect algorithm design in PRAM? (1)

C. How does the matrix-vector multiplication example illustrate strong scaling, and what would change in a weak scaling scenario? (1)

D. Why does Amdahl's Law suggest diminishing returns with increasing processors, and how does Gustafson's Law counter this? (1)

Exercise n. 2

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. In class, we studied two algorithms for minimum cut in a graph. Describe the behavior of both algorithms, including their algorithmic class and complexity. Based on the discussion held in class, explain why one might be preferable over the other. (2)

B. How do the unique table and computed table contribute to the efficiency and canonicity of BDDs?" (2)

Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Why does quicksort perform poorly in the worst case, and how can this be mitigated? Report the class of algorithms to which it belongs and its average-case complexity. According to the definition, is it an efficient algorithm? (2)

B. Describe how the insert for the Skip-list data structure works and discuss its complexity. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Why is amortized analysis useful in algorithm design, especially for data structures? (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

B. Describe the Select-ith deterministic algorithm and its complexity. Why is dividing into groups of 5 critical for the deterministic Select-ith algorithm's performance? (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

C. What does it mean if the potential difference $\Delta\Phi$ is negative in the potential method? What about the sign of Φ. (1)